

Chain and Independent Food Service Business Performance According to Income and Neighborhood in Boston



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Data Science Major Capstone

Background and Aims

- Food health and safety are essential so that illness does not spread.
- Investigating differences in food handling practices across neighborhoods helps better understand whether there are trends between socioeconomic class, neighborhood, and food safety.
- Previous study in Philadelphia and British Columbia, Canada show that chain restaurants have fewer total violations per inspection than non-chain restaurants (Tung, 2018). Non-chain restaurants generally have a higher proportion of people living in poverty, and residents live in lower income with more Black and Hispanic residents near them (Leinwand et al., 2017).
- This study is interested in the relationship between chain food inspection violations and the neighborhoods in which they reside.

Research Question

To what extent do chains perform differently (in terms of violations) than independent businesses in different neighborhoods based on income in Boston in 2024 to 2026?

Methods

- The original dataset consisted of one row per violation (*Food Establishment Inspections*, n.d.).
- National and other chains separated from independents (*Top 500 Chains*, n.d.).
- Data were spatially joined with census tracts (*TIGER/Line Shapefiles*, 2023). Coordinates of businesses were mapped to census tracts.
- Income data was then included and median income added to each row (U.S. Census Bureau, American Community Survey, 2024). There are 9 special purpose tracts and 4 tracts without income data.
- Data was then constrained to 2024-2026 and deduplicated so there is one row per inspection visit. Each visit has information with the number of violations per visit.

Data Summary

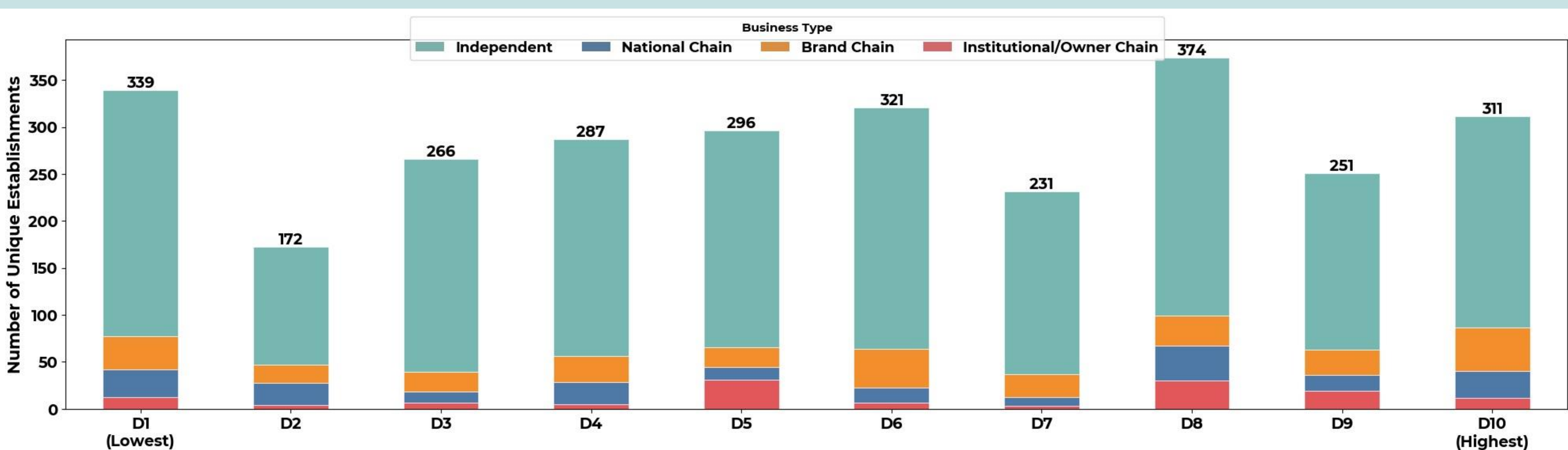


Figure 1: Unique Food Establishments by Business Type and Neighborhood Income Decile

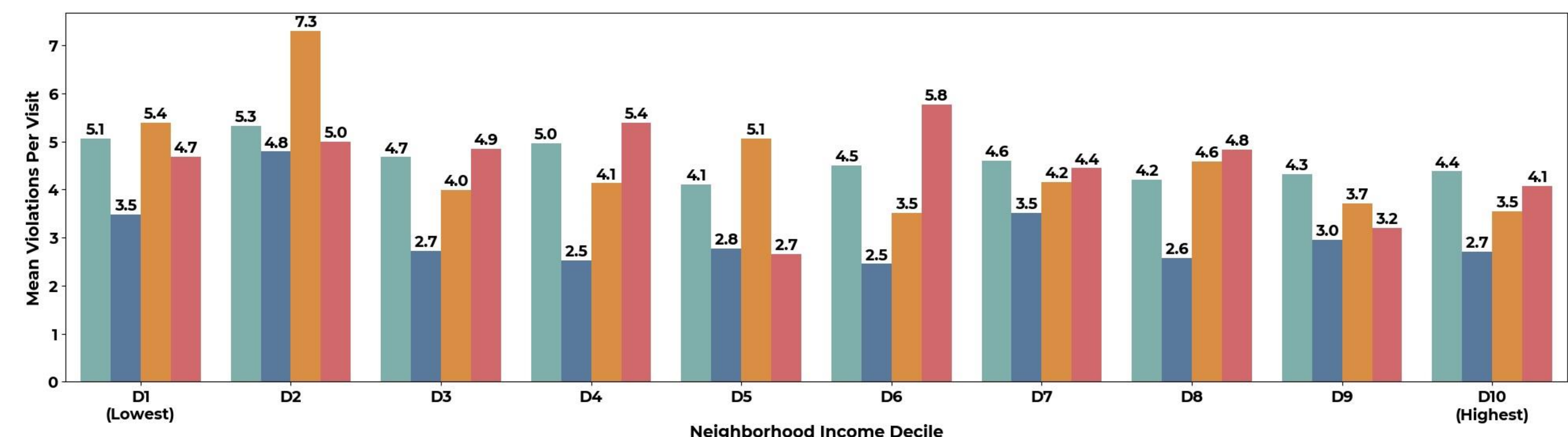


Figure 2: Mean Violations Per Visit by Business Type and Neighborhood Income Decile

Table 1: Negative Binomial Mixed Effects Interaction Model Results

Variable	Coefficient	Incidence Rate Ratio	P-Value
Intercept (baseline)	1.405586	4.08 violations per visit, baseline	<0.001
National Chain	-0.392238	32.5% fewer	<0.001
Brand Chain	0.206467	22.9% more	0.00907
Institutional/Owner Chain	-0.104277	9.9% fewer	0.38596
Median Income	-0.006885	0.7% decrease with \$10k	0.07231
National Chain x Median Income	0.001685	0.17% more with \$10k increase	0.77128
Brand Chain x Median Income	-0.017272	1.7% decrease with \$10k increase	0.00673
Institutional/Owner Chain x Median Income	0.004031	0.4% increase with \$10k increase	0.66080
Conditional R ²	0.399	Marginal R ²	0.028

Results and Discussion

- There are fewer violations in high income neighborhoods than there are in low-income neighborhoods (Figure 2, Table 1).
- Negative binomial mixed effects model was used due to overdispersion in the Poisson.
- Random effects include census tract and establishment. Fixed effects are analysis group and median income.
- Significant findings are as follows: National chains have 32.5% fewer violations than independents when income is \$0 (Table 1).
- Brand chains have 22.9% more violations than independents when income is \$0 (Table 1).
- For every \$10,000 increase in median household income, brand chain violations decrease by 1.7% (Table 1).
- Independents have 4.08 violations compared to average of 4.45 violations (Table 1).
- Fixed effects explain 2.8% of the variance (Marginal R²) and the full model explains 39.9% of the variance (conditional R²) (Table 1).

Limitations

- There is potentially temporal and spatial correlation. Two tracts next to each other may have more similarities than one far away. Time isn't accounted for in the model.
- Independents comprise most inspection visits. It is more difficult to make accurate predictions for chains (Figure 1).
- A few chains dominate the National Chain category, meaning that these few chains' outcomes represent the entire group.
- Potential clustering based on business isn't considered – a potential independence issue. One Dunkin Donuts may be more like another.

References

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- U.S. Census Bureau, American Community Survey. (2024). S1901: Income in the Past 12 Months (in 2024 Inflation-Adjusted Dollars) [Dataset]. <https://data.census.gov/all?q=S1901>